Exploring the Role of User Comments Throughout the Stages of Video-Based Task-Learning

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1. Introduction

Video-based task learning is widespread, but not personalized

- **Video** shows processes, but **a fixed viewpoint limits tacit knowledge**
- Learners **turn to comments** to fill this gap
- **Comments support:**

Questions **Goals** **Experience** **Explanations**

However...

The role of comments in the learning process is **under-explored.**

Before: Search **During: Watch** **After: Reflect**

2. Exploratory Study

Methodology

- Semi-structured interviews (~ 30 min)
- Co-watching sessions (~ 30 min)
- Analyzed using thematic analysis

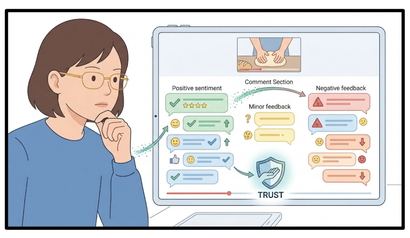


3. Findings: The Role of Comments Across Learning Stages

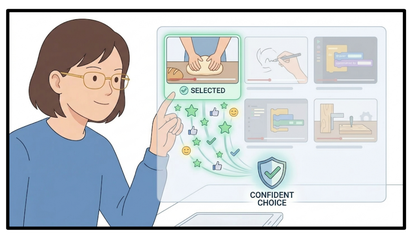
BEFORE: Trust & Video Selection

- Comments act as **trust signals** for video selection
- Comments reduce uncertainty before learning starts


Video Browsing



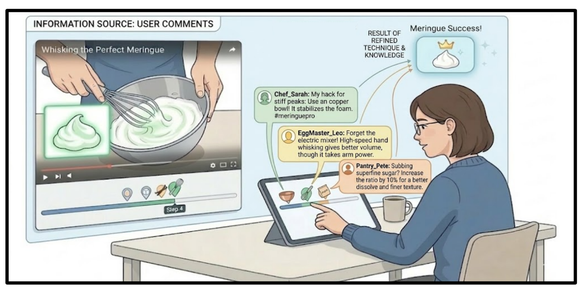
Reading Comments for Trust



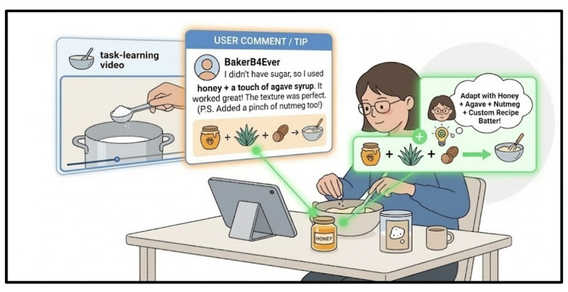
Decision Making

DURING: Sensemaking & Personalization

- Enable **collaborative understanding & adaption**
- Transform passive watching into **active learning**



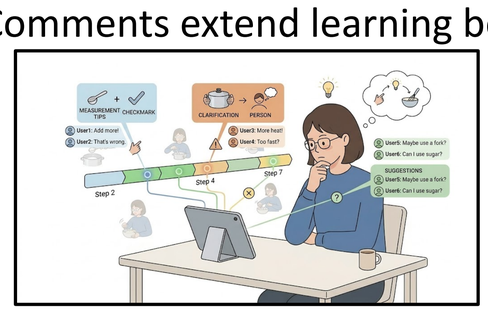
Step-specific explanation for understanding



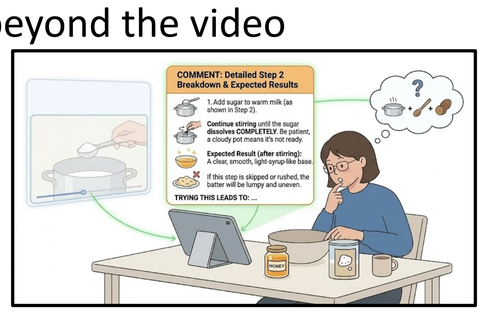
Adapt actions to users' own context

AFTER: Reflection & Extension

- Comments support **reflection and deeper learning**
- Comments extend learning beyond the video



Look back & deepen understanding



Learn more beyond the video

Surfaced Challenges

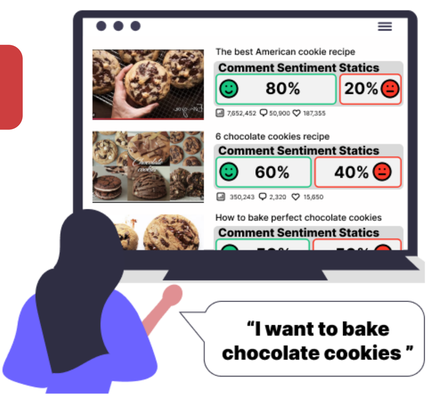
Difficult comment discovery **Disrupted task flow** **Limited access**

4. Design Opportunities

Transform **comments** from passive content into an integrated, **stage-aware learning support system**

Support informed video selection

Surface collective signals from comments.





Enable in-situ sensemaking

Embed step-based insights directly into video playback.

Reflection and knowledge extension

Structure feedback by step for effortless navigation and analysis.





Improve access & navigation

Minimize manual navigation and noise to enable seamless, hands-free mobile access.

5. Limitation & Future work

Research should explore **broader demographics and diverse platforms** to better decode the underlying motivations for social commenting.

Read the Paper